2001

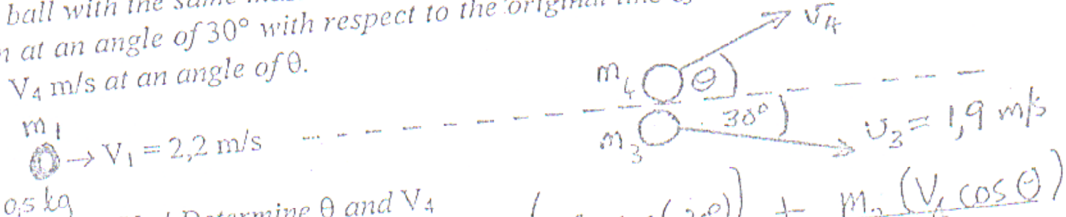
VRAAG/*QUESTION* 1

(a) Wat is die voorwaarde(s) vir 'n elastiese botsing.
When is a collision known as elastic?

A collision is elastic if both momentum and kinetic energy are conserved during the collision (2)

2

(b) 'n Bal met massa m = 0,5kg beweeg met 'n spoed V_1 = 2,2 m/s en bots teen 'n stilstaande bal met dieselfde massa m. Na die botsing beweeg een bal met 'n spoed V_3 = 1,9 m/s teen 'n hoek van 30° ten opsigte van die invalsrigting. Die ander bal beweeg teen V_4 m/s met 'n hoek θ.
A ball with mass m = 0,5 kg is moving with a speed V_1 = 2,2 m/s. It collides with a stationary ball with the same mass m. After the collision one ball moves at V_3 = 1,9 m/s in a direction at an angle of 30° with respect to the original line of motion. The other ball is moving at V_4 m/s at an angle of θ.



(i) Bepaal θ en V_4 / *Determine θ and V_4*

x-direction: $m_1 v_1 + 0 = m_1(1{,}9 \times \cos(-30°)) + m_2(V_4 \cos\theta)$

$\therefore 0{,}5(2{,}2) = 0{,}5(1{,}9\cos 30°) + 0{,}5\, V_4 \cos\theta$

$V_4 \cos\theta = 0{,}555 \quad \ldots (1)$

y-direction: $0 = m(V_4 \sin\theta) - m(1{,}9 \sin 30°)$

$0{,}5(1{,}9 \sin 30°) = 0$

$\therefore 0{,}5(V_4 \sin\theta) - 0{,}5(1{,}9\sin 30°) = 0$

$V_4 \sin\theta = 0{,}95 \quad \ldots (2)$

$(2) \div (1) \therefore \frac{V_4 \sin\theta}{V_4 \cos\theta} = \frac{0{,}95}{0{,}555} \therefore \tan\theta = 1{,}7117$

$\theta = 59{,}7°$

$\therefore V_4 = \frac{0{,}555}{\cos\theta} = 1{,}1 \text{ m/s}$ (9)

9

(ii) Bepaal of hierdie 'n elastiese of nie-elastiese botsing was.
Determine if this collision was elastic or inelastic.

$KE_{before} = \frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2}(0{,}5)(2{,}2)^2 = 1{,}21 \text{ J}$

$KE_{after} = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2 = \frac{1}{2}(0{,}5)(1{,}9)^2 + \frac{1}{2}(0{,}5)(1{,}1)^2$

$= 0{,}9025 + 0{,}3025$ (4)

Vraag / *Question* 1(b)(iii)(Vervolg)(*Continue*) / .....

$= 1{,}21 \text{ J}$

$\therefore KE_{before} = KE_{after} \therefore$ It is an elastic collision